

Contents

Awesome Mnemonics - Complete Guide	2
Table of Contents	2
Quick Reference & On-Call Guide	3
Categorized Index	3
Ops	3
RCA (Root Cause Analysis)	4
Systems Design	4
Human Factors	4
Personal Life	5
Problem Solving Techniques	5
PREPARE	5
PADDER	5
ICEBERG	6
IDEA	6
5 Whys	6
8D Approach	7
5Ps	7
Problem Analysis	7
RACI	7
PESTEL	8
SWOT	8
SET (Systems Engineering Triangle)	8
Problem Resolution Threats	9
DICE	9
FATE	9
PEST	9
Communication & Conflict	9
BREATHE	9
PAUSE	10
WAIT	10
Stress & Resilience	10
PACE	10
STOP	10
ARIES	11
HELP	11
HANDLE	11
CALM	11
PUSH	12
HOPE	12
SHINE	12
Infrastructure & Systems Engineering	12
TRACE (Network Troubleshooting)	12
SCALE (Infrastructure Design)	12
DEBUG (Code & System Analysis)	13
Proven Mnemonic Pipelines	13
1. CRISIS RESPONSE CHAIN	13
2. CONFLICT RESOLUTION CHAIN	13
3. ROOT CAUSE INVESTIGATION CHAIN	14
4. STRATEGIC DESIGN CHAIN	14
5. STRESS BURNOUT RECOVERY CHAIN	15
6. RAPID TRIAGE CHAIN	16
Pipeline Selection Matrix	16

Pro Tips for Using Pipelines	16
Common Mistakes to Avoid	17
Jumping to Solutions	17
Using Wrong Pipeline	17
Incomplete Execution	17
Solo Hero Mode	17
Analysis Paralysis	17

Awesome Mnemonics - Complete Guide

Memory hacks that compress complex workflows into actionable acronyms. Use during incidents, RCAs, design decisions, or high-stress situations.

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Table of Contents

- Quick Reference & On-Call Guide
- Categorized Index
 - Ops
 - RCA (Root Cause Analysis)
 - Systems Design
 - Human Factors
 - Personal Life
- Problem Solving Techniques
 - PREPARE
 - PADDER
 - ICEBERG
 - IDEA
 - 5 Whys
 - 8D Approach
 - 5Ps
- Problem Analysis
 - RACI
 - PESTEL
 - SWOT
 - SET (Systems Engineering Triangle)
- Problem Resolution Threats
 - DICE
 - FATE
 - PEST
- Communication & Conflict
 - BREATHE
 - PAUSE
 - WAIT
- Stress & Resilience
 - PACE
 - STOP
 - ARIES
 - HELP
 - HANDLE
 - CALM
 - PUSH

- HOPE
- SHINE
- Infrastructure & Systems Engineering
 - TRACE (Network Troubleshooting)
 - SCALE (Infrastructure Design)
 - DEBUG (Code & System Analysis)
- Proven Mnemonic Pipelines
 - Crisis Response Chain
 - Conflict Resolution Chain
 - Root Cause Investigation Chain
 - Strategic Design Chain
 - Stress Burnout Recovery Chain
 - Rapid Triage Chain
- Pipeline Selection Matrix
- Pro Tips for Using Pipelines
- Common Mistakes to Avoid

Quick Reference & On-Call Guide

Production incident? → Crisis Response Chain (STOP → TRACE → DEBUG → 8D)

Need quick decision? → Rapid Triage Chain (IDEA → DICE → FATE)

Situation	Use This	Time
Immediate crisis	IDEA → STOP	2-5 min
Root cause needed	ICEBERG + 5 Whys	30-60 min
Team conflict	BREATHE → PAUSE → WAIT	5-10 min
Strategic planning	PREPARE + SWOT	1-2 hours
System design	SCALE + PESTEL	Planning
Technical debugging	TRACE → DEBUG	Variable
Stress overload	PACE → ARIES → CALM	10-15 min

Categorized Index

Quick lookup by functional area with “when to use” guidance

Ops

Incident response, troubleshooting, and operational decision-making

Mnemonic	When to Use
STOP	First response to any crisis—immediate stress management (2-5 min)
IDEA	Quick, simple problems requiring fast action (2-5 min)
TRACE	Network/system diagnostics—start with connectivity tests
DEBUG	Code or system analysis when problem definition is unclear
DICE	Identify blockers (Delay, Incompetence, Conflict, External factors) before execution
FATE	Validate resource availability (Funding, Allocation, Time, Expertise) for feasibility

Mnemonic	When to Use
PEST	Identify external threats (Political, Economic, Social, Technological) to solutions

RCA (Root Cause Analysis)

Deep investigation, recurring issues, and prevention

Mnemonic	When to Use
ICEBERG	Complex problems with hidden root causes (30-60 min)
5 Whys	Drill down to root cause—keep asking until systemic failure is found
8D Approach	Critical incidents requiring formal resolution and prevention
PADDER	Data-driven problem solving when patterns need identification
PREPARE	Strategic planning for medium complexity problems (1-2 hours)

Systems Design

Architecture, planning, and technical decision-making

Mnemonic	When to Use
SCALE	Infrastructure design—Security, Capacity, Automation, Load balancing, Error handling
SWOT	Evaluate options—Strengths, Weaknesses, Opportunities, Threats (1-2 hours)
PESTEL	Strategic planning considering external factors (Political, Economic, Sociocultural, Technological, Environmental, Legal)
SET	Set stakeholder expectations—pick 2: Good, Fast, Cheap

Human Factors

Team dynamics, conflict resolution, and collaboration

Mnemonic	When to Use
RACI	Resolve role confusion—assign Responsible, Accountable, Consulted, Informed
WAIT	Before speaking—ask “Why am I talking?” to choose the right response
BREATHE	First step when tensions rise—regulate emotions before responding
PAUSE	When BREATHE isn’t enough—step away and evaluate before acting (5-20 min)
DICE	Check for Conflict as a blocker in team execution
FATE	Validate team Expertise and resource Allocation

Personal Life

Stress management, burnout recovery, and personal resilience

Mnemonic	When to Use
PACE	Immediate stress management—Physical activity, Avoid unhealthy behaviors, Coping skills, Emotional awareness
ARIES	Lifestyle changes for long-term stress reduction (2-4 weeks)
CALM	Build resilience—Confidence, Awareness, Logic, Mindfulness
HELP	Stress management strategies—Handle one at a time, Exercise, Learn to relax, Pace yourself
HANDLE	Six stress-management strategies including positive attitude and support system
PUSH	Build positivity—Practice Gratitude, Use affirmations, Spend time with positive people, Have positive outlook
HOPE	Harness positive thoughts, Open up, Prioritize self-care, Exercise regularly
SHINE	Ongoing practice for sustainable positivity—Stay present, Healthy perspective, Identify positive activities, Nourish relationships, Express yourself

Problem Solving Techniques

Start here for systematic approaches to complex problems

PREPARE

P - Prioritize the problem
R - Research & brainstorm solutions
E - Evaluate available options
P - Plan steps to resolve issue
A - Act on the plan
R - Reflect on results
E - Evaluate and revise plan as necessary

When to use: - Strategic planning sessions (1-2 hours) - Medium complexity problems with multiple stakeholders - When you need structured documentation

Common pitfalls: - **Analysis paralysis** - Spending 2 hours planning a 10-minute problem. If you're past 30 minutes on "Research", switch to IDEA. - **Skipping reflection** - Acting without the final "Reflect/Evaluate" creates recurring issues. Always close the loop. - **Wrong tool for crisis** - Don't PREPARE during a production outage. Use STOP → TRACE → DEBUG first, PREPARE during post-mortem.

Combines well with: RACI (assign responsibilities), SWOT (evaluate options), ICEBERG (if reflection reveals deeper issues)

Real-world example: *Planning infrastructure migration - Use PREPARE to structure approach, RACI during planning phase for role clarity, SWOT to evaluate cloud provider options*

PADDER

P - Pinpoint problem
A - Analyze data and look for patterns
D - Develop solution & consider other ways to solve the issue - try to have more than one

D - Design action plan
E - Execute action plan & Monitor Results
R - Reevaluate and refine plan as needed

When to use: - Data-driven problem solving - When patterns need to be identified - Pairs with 8D Approach's D3 (Interim Containment) for quick fixes

Combines well with: IDEA (simpler version), 8D Approach (formal resolution)

Real-world example: *Recurring server crashes - Pinpoint timing, Analyze logs for patterns, Develop interim solutions (restart service) + permanent fix (increase memory), Monitor effectiveness*

ICEBERG

I - Identify issue(s)
C - Collect data and analyze situation
E - Examine possible (root) causes
B - Brainstorm solutions
E - Execute solution(s)
R - Review, evaluate, and adjust solutions
G - Gather feedback

When to use: - Complex problems requiring deep analysis (30-60 min) - When surface symptoms hide deeper root causes - Escalate from IDEA when complexity increases

Common pitfalls: - **Going too deep on simple problems** - Using ICEBERG for a 5-minute password reset. Start with IDEA, escalate only if complexity emerges. - **Skipping feedback (G)** - Gathering feedback seems optional but prevents recurrence. Always complete the full cycle.

Combines well with: 5 Whys (deeper root cause), 8D Approach (formal prevention), IDEA (start simple, escalate if needed)

Real-world example: *Network performance degradation - Identify slowness, Collect metrics (latency, packet loss), Examine causes (routing changes, bandwidth saturation), Brainstorm solutions, Execute, Review with team, Gather feedback from users*

IDEA

I - Identify problem
D - Develop Solution
E - Execute Solution
A - Assess Solution

When to use: - Quick, simple problems (2-5 minutes) - Crisis situations requiring immediate action - Use STOP first if under stress

Common pitfalls: - **Escalation failure** - Not moving to ICEBERG or PREPARE when problem reveals complexity. If you're past 10 minutes or hitting multiple blockers, escalate. - **Skipping assessment** - Executing without validating the solution worked. The "A" step prevents recurring issues.

Combines well with: STOP (crisis stress management), PREPARE/ICEBERG (escalation path for complex issues)

Real-world example: *User can't access VPN - Identify (connection error), Develop (reset credentials), Execute (send new password), Assess (confirm access restored)*

5 Whys

- Keep asking why till root causes are identified

When to use: - Essential for root cause analysis - During 8D Approach's D4 (Root Cause Analysis) - Combine with ICEBERG for systematic deep dives

Common pitfalls: - **Stopping at symptoms** - Stopping at "Why #3: Database slow" instead of drilling to root cause (missing migration). Keep asking until you reach a process/systemic failure. - **Assuming single root cause** - Complex problems often have multiple root causes. Use 5 Whys for each branch.

Combines well with: ICEBERG (structured approach), 8D Approach (formal incident management)

Real-world example: *Deployment failures - Why? Pipeline failed. Why? Tests timed out. Why? Database slow. Why? Index missing. Why? Schema change didn't include migration. Root cause: Missing migration validation step*

8D Approach

Industry standard in automotive/manufacturing, adapted for IT incident management

- D1 - Form a team
- D2 - Describe the problem
- D3 - Interim Containment Action (the "band-aid")
- D4 - Root Cause Analysis & Escape Point(s)
- D5 - Permanent Corrective Actions
- D6 - Implement & Validate Corrective Actions
- D7 - Prevent reoccurrence(s)
- D8 - Closure & Celebration

When to use: - Critical incidents requiring formal resolution - Problems needing documentation and prevention - When team coordination is essential (D1: use RACI)

Common pitfalls: - **Stopping at D3** - Implementing the band-aid but never reaching D7 (Prevent Reoccurrence) means it will happen again. - **Solo 8D** - Trying to do all 8 steps alone instead of D1 (Form a team). Use RACI during D1 to clarify roles. - **Bureaucracy creep** - Over-formalizing 8D for simple problems. Use IDEA or PREPARE for non-critical issues.

Combines well with: Padder (D3 interim fixes), 5 Whys + ICEBERG (D4 root cause), RACI (D1 team formation)

Real-world example: *Data breach incident - Form security response team (RACI roles), Describe scope, Contain (disable compromised accounts), Analyze root cause (5 Whys: phishing → no MFA → insufficient training), Implement MFA, Validate with penetration test, Prevent (mandatory security awareness), Celebrate team response*

5Ps

- Poor planning produces pitiful products

Problem Analysis

Frameworks for understanding scope and impact

RACI

- Used to identify the roles and responsibilities of different stakeholders in a problem-solving process

- R - Responsible (does the work)
- A - Accountable (final approval)
- C - Consulted (provides input)
- I - Informed (kept updated)

When to use: - Resolving role confusion in teams - 8D Approach's D1 (Form a team) step - End of WAIT → BREATHE → PAUSE chain for conflict resolution

Common pitfalls: - **Multiple Accountables** - More than one "A" creates confusion. There should be exactly one Accountable person per task. - **Too many Consulted** - Adding everyone as "C" slows decisions. Be selective - only include those with critical input.

Real-world example: *Infrastructure upgrade project - Responsible: DevOps engineers, Accountable: Infrastructure Manager, Consulted: Security team, Informed: All developers*

PESTEL

- Used to identify and analyze the external factors that may impact a problem or decision

P - Political
E - Economic
S - Sociocultural
T - Technological
E - Environmental
L - Legal

When to use: - Strategic planning external factor analysis - SCALE infrastructure design validation - Architecture reviews considering compliance/regulations

Real-world example: *Cloud migration planning - Political (vendor lock-in concerns), Economic (cost optimization), Technological (API compatibility), Legal (data sovereignty requirements)*

SWOT

- Used to identify and analyze the internal and external factors that may impact a problem or decision

S - Strengths
W - Weaknesses
O - Opportunities
T - Threats

When to use: - PREPARE's "Evaluate options" step - SCALE design validation - Strategic decision making (1-2 hours)

Real-world example: *Choosing deployment strategy - Strengths: automated rollback, Weaknesses: longer deployment time, Opportunities: canary testing, Threats: increased complexity*

SET (Systems Engineering Triangle)

- Also called "Project Management Triangle" or "Iron Triangle"

1. Draw a triangle
2. Put one of these 3 words at each corner: "good", "fast", "cheap"
3. Pick 2 of them
4. The other word is what the solution will not likely be
 - A fast, cheap solution will not likely be good
 - A fast and good solution will not likely be cheap
 - A good and cheap solution will not likely be fast

When to use: - Setting stakeholder expectations - End of SCALE → SWOT → PESTEL chain - Architecture trade-off discussions

Real-world example: *Urgent security patch needed - Choose: Good + Fast = Not Cheap (overtime, additional resources). Manage expectations with leadership accordingly*

From the SeBok - Systems Engineering Body of Knowledge

Problem Resolution Threats

Identify blockers before they derail your solution

DICE

D - Delay
I - Incompetence
C - Conflict
E - External factors

When to use: - Part of IDEA → DICE → FATE rapid triage chain - Identifying blockers before execution - Risk assessment in PREPARE phase

FATE

F - Funding
A - Allocation of resources
T - Time
E - Expertise

When to use: - Resource validation in rapid triage - Feasibility assessment (10-30 min) - After DICE to validate resource availability

PEST

P - Political
E - Economic
S - Social
T - Technological

When to use: - Identifying external threats to solutions - Consider ways to combat/remove these threats

Communication & Conflict

Stay calm and productive during difficult conversations

BREATHE

B - Breathe deeply and slowly
R - Remain rational and listen
E - Empathize with the other person's problem
A - Ask questions to understand
T - Take a break if needed
H - Hold back from reacting
E - Express yourself calmly

When to use: - First step when tensions rise (breathing regulates emotions) - Part of WAIT → BREATHE → PAUSE → RACI conflict resolution chain - If break needed, escalate to PAUSE

Common pitfalls: - **Fake composure** - Going through motions without actually regulating emotions. The breathing must be intentional and deep. - **Weaponizing calm** - Using “Let’s BREATHE” dismissively to avoid addressing concerns. This damages trust - use it genuinely.

Combines well with: WAIT (listen first), PAUSE (step away if needed), STOP (stress management)

Real-world example: *Stakeholder disagrees with technical approach in meeting - Breathe deeply (regulate emotions), Remain rational, Empathize with their concerns, Ask clarifying questions, Take 5-minute break if tension escalates, Hold back defensive reactions, Express technical rationale calmly*

PAUSE

P - Put things in perspective
A - Acknowledge your feelings and theirs
U - Understand that you don't have to act/react right away
S - Step Away from the situation
E - Evaluate options and plan before acting

When to use: - When BREATHE isn't enough - need physical separation - "Step Away" connects to WAIT philosophy - Combine with STOP for immediate stress de-escalation (5-20 min)

Combines well with: BREATHE (first step), WAIT (listen before speaking), STOP (stress response)

Real-world example: *Heated debate about architecture decision - Put in perspective (not life-or-death), Acknowledge both viewpoints have merit, Don't decide now, Step away for lunch break, Evaluate pros/cons offline, Return with structured comparison*

WAIT

- "Why am I troubled / talking?"
- Not all things need an answer or response; don't seek trouble and don't answer negativity negatively

When to use: - Ask yourself before speaking - often listening is better - First step in WAIT → BREATHE → PAUSE → RACI chain - Use BREATHE first to regulate, then WAIT to choose response

Common pitfalls: - **Passive-aggressive silence** - Using WAIT to avoid necessary communication. This isn't "don't respond" - it's "choose the right response." - **Escalation avoidance** - Not speaking up when issues need addressing. WAIT helps you respond effectively, not disappear.

Combines well with: BREATHE (emotional regulation), PAUSE (step away technique)

Real-world example: *Email with accusatory tone arrives - Ask "Why am I troubled?" (ego/defensiveness), "Why am I talking?" (to defend or to resolve?), Choose not to respond immediately, Use BREATHE + PAUSE, Respond later with facts, not emotion*

Stress & Resilience

Staying calm and connected during high-pressure work

PACE

P - Physical activity
A - Avoiding unhealthy behaviors
C - Coping skills
E - Emotional awareness

When to use: - First step in PACE → ARIES → CALM → SHINE burnout recovery chain - Immediate stress management actions

STOP

S - Step back
T - Take a deep breath

O - Observe what is happening
P - Pull back and put things in perspective

When to use: - First response to ANY crisis - immediate stress management (2-5 min) - Start of STOP → TRACE → DEBUG → 8D crisis response chain - Use with IDEA for quick problem resolution in crisis

Common pitfalls: - **Going through motions** - Rushing through STOP without actually calming. The “Take a deep breath” must be intentional - pause for 3-5 seconds. - **Stopping at STOP** - Using STOP but not proceeding to next step (TRACE/IDEA). STOP is the foundation, not the solution.

Combines well with: BREATHE + PAUSE (breathing techniques), IDEA (quick problem solving)

Real-world example: *Production alert at 2 AM - Step back (don't panic), Take deep breath (reduce adrenaline), Observe (read alert details), Pull back perspective (assess severity before waking team), Then proceed to TRACE for diagnostics*

ARIES

A - Avoid unnecessary stress
R - Relax and take breaks
I - Incorporate physical activity into your routine
E - Eat a healthy diet
S - Sleep well

When to use: - Lifestyle changes in PACE → ARIES → CALM chain - Long-term stress reduction (2-4 weeks)

HELP

- This mnemonic can help you remember four ways to manage stress:

H - Handle problems one at a time
E - Exercise regularly
L - Learn to relax
P - Pace yourself and set limits.

HANDLE

- This mnemonic can help you remember six stress-management strategies:

H - Have a positive attitude
A - Avoid unnecessary stress
N - Nurture a strong support system
D - Deal with problems directly
L - Learn to relax,
E - Exercise regularly

CALM

C - Confidence: Believe in your abilities and strengths
A - Awareness: Stay conscious of your thoughts and feelings
L - Logic: Use rational thinking to overcome doubts
M - Mindfulness: Practice being present and focused

When to use: - Builds long-term resilience and confidence - Confidence connects to HOPE and SHINE (both build positive self-image) - Use after PACE or ARIES for comprehensive stress management

Combines well with: Positivity (HOPE, SHINE), Stress Management (PACE, ARIES)

PUSH

P - Practice Gratitude
U - Use positive affirmations
S - Spend time with positive people
H - Have a positive outlook

HOPE

H - Harness positive thoughts
O - Open up to others
P - Prioritize self-care
E - Exercise regularly

SHINE

S - Stay present, in the moment
H - Have a healthy positive perspective
I - Identify and do positive activities
N - Nourish positive relationships
E - Express yourself

When to use: - End of PACE → ARIES → CALM → SHINE burnout recovery chain - Ongoing practice for sustainable positivity - Connects to CALM for building positive self-image

Infrastructure & Systems Engineering

Technical mnemonics for infrastructure and DevOps work

TRACE (Network Troubleshooting)

T - Test connectivity (ping, traceroute)
R - Review logs and metrics
A - Analyze packet captures
C - Check configurations
E - Escalate with documented evidence

When to use: - Use with PREPARE and 8D Approach for systematic resolution - Start with T (Test) for quick diagnostics, escalate to A (Analyze) for deep dives - Document findings at each step for 8D Approach's D2 (Describe the problem)

Combines well with: Problem Solving (PREPARE, 8D Approach)

SCALE (Infrastructure Design)

S - Security by design
C - Capacity planning
A - Automation-first
L - Load balancing
E - Error handling/resilience

When to use: - Use SWOT analysis alongside for design validation - Apply PESTEL to evaluate external factors affecting infrastructure - Consider SET Triangle trade-offs (fast/cheap/good) for each component

Combines well with: Problem Analysis (SWOT, PESTEL, SET)

DEBUG (Code & System Analysis)

D - Define the problem (what changed?)
E - Examine error messages/logs
B - Break down into components
U - Understand data flow
G - Generate hypothesis and test

When to use: - Combines 5 Whys questioning with ICEBERG's systematic approach - Start with D (Define)
- often the problem is unclear or misstated - Use with TRACE for network/system issues

Combines well with: Problem Solving (5 Whys, ICEBERG), Infrastructure (TRACE)

Proven Mnemonic Pipelines

When certain mnemonics are combined in sequence, they create powerful workflows that amplify effectiveness. These chains are battle-tested for specific high-pressure scenarios.

1. CRISIS RESPONSE CHAIN

STOP → TRACE → DEBUG → 8D

When to use: Production outages, system failures, critical incidents

The Flow:

S - Step back (STOP)
T - Take a deep breath (STOP)
O - Observe what is happening (STOP)
P - Pull back and put things in perspective (STOP)
↓
T - Test connectivity (TRACE)
R - Review logs and metrics (TRACE)
A - Analyze packet captures (TRACE)
C - Check configurations (TRACE)
E - Escalate with documented evidence (TRACE)
↓
D - Define the problem (DEBUG)
E - Examine error messages/logs (DEBUG)
B - Break down into components (DEBUG)
U - Understand data flow (DEBUG)
G - Generate hypothesis and test (DEBUG)
↓
D1-D8 - Full 8D resolution cycle (8D Approach)

Why it works: STOP stabilizes you emotionally, TRACE gives you diagnostic data, DEBUG structures your analysis, 8D ensures you prevent recurrence.

Time investment: 1-4 hours (depending on complexity)

2. CONFLICT RESOLUTION CHAIN

WAIT → BREATHE → PAUSE → RACI

When to use: Team disagreements, tense meetings, stakeholder conflicts

The Flow:

W - "Why am I talking?" (WAIT - listen first)
 ↓
 B - Breathe deeply and slowly (BREATHE)
 R - Remain rational and listen (BREATHE)
 E - Empathize with the other person's problem (BREATHE)
 A - Ask questions to understand (BREATHE)
 ↓
 P - Put things in perspective (PAUSE)
 A - Acknowledge your feelings and theirs (PAUSE)
 U - Understand you don't have to react immediately (PAUSE)
 S - Step Away from the situation (PAUSE)
 E - Evaluate options and plan (PAUSE)
 ↓
 R - Responsible (RACI - clarify roles)
 A - Accountable (RACI)
 C - Consulted (RACI)
 I - Informed (RACI)

Why it works: WAIT prevents escalation, BREATHE regulates emotions, PAUSE creates space for rational thought, RACI resolves role confusion (often the root cause).

Time investment: 5-20 minutes

3. ROOT CAUSE INVESTIGATION CHAIN

ICEBERG → 5 Whys → PADDER → RACI

When to use: Recurring issues, complex system problems, post-incident analysis

The Flow:

I - Identify issue(s) (ICEBERG)
 C - Collect data and analyze situation (ICEBERG)
 E - Examine possible root causes (ICEBERG)
 ↓
 Why #1 → Why #2 → Why #3 → Why #4 → Why #5 (5 Whys)
 ↓
 D - Develop solution (PADDER)
 D - Design action plan (PADDER)
 E - Execute action plan & Monitor (PADDER)
 R - Reevaluate and refine (PADDER)
 ↓
 R/A/C/I - Assign responsibilities (RACI)

Why it works: ICEBERG provides structure, 5 Whys drills to root cause, PADDER creates action plan, RACI ensures accountability.

Time investment: 1-2 hours

4. STRATEGIC DESIGN CHAIN

SCALE → SWOT → PESTEL → SET

When to use: Infrastructure planning, architecture reviews, capacity planning

The Flow:

S - Security by design (SCALE)
 C - Capacity planning (SCALE)
 A - Automation-first (SCALE)
 L - Load balancing (SCALE)
 E - Error handling/resilience (SCALE)
 ↓
 S - Strengths (SWOT - evaluate design)
 W - Weaknesses (SWOT)
 O - Opportunities (SWOT)
 T - Threats (SWOT)
 ↓
 P - Political (PESTEL - external factors)
 E - Economic (PESTEL)
 S - Sociocultural (PESTEL)
 T - Technological (PESTEL)
 E - Environmental (PESTEL)
 L - Legal (PESTEL)
 ↓

Pick 2: Good / Fast / Cheap (SET Triangle - set expectations)

Why it works: SCALE sets technical requirements, SWOT evaluates approach, PESTEL identifies external risks, SET manages stakeholder expectations.

Time investment: 2-4 hours (planning phase)

5. STRESS BURNOUT RECOVERY CHAIN

PACE → ARIES → CALM → SHINE

When to use: Long-term stress, approaching burnout, need for lifestyle reset

The Flow:

P - Physical activity (PACE - immediate actions)
 A - Avoiding unhealthy behaviors (PACE)
 C - Coping skills (PACE)
 E - Emotional awareness (PACE)
 ↓
 A - Avoid unnecessary stress (ARIES - lifestyle changes)
 R - Relax and take breaks (ARIES)
 I - Incorporate physical activity (ARIES)
 E - Eat a healthy diet (ARIES)
 S - Sleep well (ARIES)
 ↓
 C - Confidence (CALM - mental framework)
 A - Awareness (CALM)
 L - Logic (CALM)
 M - Mindfulness (CALM)
 ↓
 S - Stay present (SHINE - ongoing practice)
 H - Have healthy perspective (SHINE)
 I - Identify positive activities (SHINE)
 N - Nourish relationships (SHINE)
 E - Express yourself (SHINE)

Why it works: PACE handles immediate stress, ARIES addresses root lifestyle causes, CALM builds mental resilience, SHINE creates sustainable positivity.

Time investment: Ongoing (2-4 weeks for habit formation)

6. RAPID TRIAGE CHAIN

IDEA → DICE → FATE

When to use: Quick wins needed, time-critical decisions, assessing feasibility

The Flow:

I - Identify problem (IDEA)
D - Develop Solution (IDEA)
E - Execute Solution (IDEA)
A - Assess Solution (IDEA)
↓
D - Delay (DICE - check for blockers)
I - Incompetence (DICE)
C - Conflict (DICE)
E - External factors (DICE)
↓
F - Funding (FATE - resource check)
A - Allocation of resources (FATE)
T - Time (FATE)
E - Expertise (FATE)

Why it works: IDEA provides fast framework, DICE identifies blockers, FATE validates resource availability.

Time investment: 10-30 minutes

Pipeline Selection Matrix

Your Situation	Pipeline	Key Benefit
System down NOW	STOP → TRACE → DEBUG → 8D	Systematic crisis response
Team conflict escalating	WAIT → BREATHE → PAUSE → RACI	De-escalation + role clarity
Same issue keeps happening	ICEBERG → 5 Whys → PADDER → RACI	Deep root cause + prevention
Designing new infrastructure	SCALE → SWOT → PESTEL → SET	Complete planning framework
Feeling burned out	PACE → ARIES → CALM → SHINE	Comprehensive stress recovery
Need quick decision	IDEA → DICE → FATE	Fast feasibility assessment

Pro Tips for Using Pipelines

1. **Don't skip steps** - Each mnemonic builds on the previous one
2. **Document at each stage** - Your STOP insights inform TRACE, TRACE data feeds DEBUG, etc.

3. **Know when to branch** - If IDEA reveals complexity, switch to ICEBERG → 5 Whys
 4. **Combine with stress tools** - Always start with STOP if you're emotionally activated
 5. **Teach your team** - Shared vocabulary accelerates collaboration
-

Common Mistakes to Avoid

Jumping to Solutions

Problem: Skipping STOP or WAIT when stressed → poor decisions

Fix: Always use STOP first in crises, WAIT before reacting in conflicts

Example: *Production alert at 2 AM - Jumping straight to SSH without STOP → panic-driven wrong server reboot → extended downtime. Instead: STOP (30 seconds) → assess severity → then TRACE*

Using Wrong Pipeline

Problem: IDEA won't solve systemic issues

Fix: Use ICEBERG → 5 Whys for recurring problems

Example: *Database timeouts happen weekly - Using IDEA to restart service each time → problem returns. Instead: ICEBERG → 5 Whys reveals missing connection pool config → permanent fix*

Incomplete Execution

Problem: Starting 8D Approach but stopping at D3 (band-aid)

Fix: Always reach D7 (Prevent Reoccurrence) or the problem returns

Solo Hero Mode

Problem: Forgetting RACI → no accountability when you're unavailable

Fix: Use RACI in 8D Approach's D1 (Form a team)

Analysis Paralysis

Problem: PREPARE → ICEBERG → 5 Whys → 8D for simple problems

Fix: Start with IDEA, escalate only if complexity emerges

For the latest updates and contributions, visit: <https://github.com/StewAlexander-com/Awesome-Mnemonics>